

Monte Carlo and Crank–Nicolson option pricing

Numerical methods and an option-chain application

A comparison of simulation and finite-difference methods for option values and hedge ratios, followed by a market-data application with pricing and risk diagnostics.

Methods	C++; Euler–Maruyama; Crank–Nicolson; Black–Scholes; Python visualisation
Data	Specified CEV-type model; supplied SPY option-chain snapshot, 18 February 2026
Revision	21 September 2026

1 Executive summary

This project compares Monte Carlo simulation and Crank–Nicolson finite differences for a European put under a CEV-type diffusion, then applies the numerical methods to a supplied option-chain snapshot.

In the specified diffusion, Monte Carlo gives a put value of 9.0577 with a 95% sampling interval of [8.7989, 9.3165]. Crank–Nicolson gives 9.1295. The corresponding deltas are -0.4009 and -0.4071 , with the finite-difference result inside the Monte Carlo delta interval. An effective-volatility Black–Scholes value of 9.1492 provides an approximate reference, not an exact solution to the CEV-type model.

The market application covers 947 contracts across 10 expiries. Closed-form Black–Scholes and Crank–Nicolson give similar aggregate prices, while comparisons with quoted mids reveal systematic differences that sampling noise alone does not explain. Volatility, Greeks and scenario plots connect the numerical work to risk interpretation.

The supplied snapshot identifies SPY and Yahoo Finance via yfinance. This corrects the earlier Bloomberg/SPX description; the legacy figure filenames are retained. The exercise demonstrates numerical consistency and model diagnostics, not independently calibrated market fair values or arbitrage opportunities.

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2 Data and scope

2.1 Project origin and scope

This work adapts a project completed during the MSc in Financial Mathematics at King’s College London. The finite-difference and Monte Carlo methods build on John Armstrong’s Computational and Numerical Methods in Mathematical Finance lectures [1].

Part A compares numerical methods under a specified CEV-type model. Part B applies European Black–Scholes, Monte Carlo and PDE calculations to an option-chain snapshot. These are distinct modelling settings: the effective-volatility reference in Part A is approximate, whereas Black–Scholes is an exact closed-form reference for Part B’s assumed European GBM model.

2.2 Specified diffusion: Part A

It is given that a stock price S_t evolves according to

$$dS_t = \mu S_t dt + \sigma(t) S_t^\gamma dW_t.$$

However, for pricing we work under the risk-neutral measure $\mathbb{Q}$, where the drift becomes the risk-free rate r (under standard no-arbitrage assumptions). Hence the dynamics we use is:

$$dS_t = r S_t dt + \sigma(t) S_t^\gamma dW_t,$$

where W_t denotes a $\mathbb{Q}$ -Brownian motion.

The parameters are $\gamma = 0.95$, $\sigma(t) = 0.18 + 0.07 \cdot \frac{t}{T}$, $K = 140$, $r = 0.02$, $S_0 = 140$, and $T = 1.40$.

The goal in Part A is to compute the European put price $V(S, t)$ and delta Δ under the risk-neutral dynamics, then compare two numerical approaches:

- Monte Carlo simulation using Euler–Maruyama,
- Crank–Nicolson finite differences applied to the pricing PDE.

To benchmark the outputs, we also compute an *estimated* Black–Scholes value using an effective volatility σ_{eff} , since the true model here is not log-normal.

Part B uses market strikes, expiries, quotes and implied volatilities as inputs to a log-normal European pricing model with continuous dividend yield. The purpose is to compare numerical implementations and inspect model–market differences.

2.3 Market snapshot: Part B

The supplied project files `snapshot.csv` and `options_chain_clean.csv` identify the underlying as SPY and the source as Yahoo Finance via `yfinance`. The snapshot is timestamped 18 February 2026 at 02:59:31 UTC and records spot 682.85 and dividend yield 1.050%. Its 947 contracts, 280 calls and 667 puts match the report’s chain summary.

The source description is therefore SPY/Yahoo Finance. The original BBG figure filenames are retained for compatibility and do not establish Bloomberg provenance. Prices are quoted in the supplied USD price units. Contract exercise, dividend and quote-timing conventions need to be reconciled before interpreting differences from market mids as valuation errors.

3 Methodology

3.1 Pricing PDE and hedging argument

Consider the option value $V(S, t)$ and form a self-financing hedged portfolio by shorting one option and holding Δ shares:

$$\Pi(t) = -V(S_t, t) + \Delta_t S_t.$$

The portfolio change is

$$d\Pi = -dV + \Delta dS.$$

Using Itô's lemma on $V(S_t, t)$ under the risk-neutral dynamics:

$$dV = V_t dt + V_S dS + \frac{1}{2} V_{SS} (dS)^2.$$

Since $dS = rS dt + \sigma(t)S^\gamma dW$, we have $(dS)^2 = \sigma^2(t)S^{2\gamma} dt$ (and the mixed terms vanish). Substituting gives:

$$dV = \left(V_t + rSV_S + \frac{1}{2} \sigma^2(t)S^{2\gamma} V_{SS} \right) dt + \sigma(t)S^\gamma V_S dW.$$

Now plug this into the portfolio dynamics. If we choose the hedge ratio $\Delta = V_S$, the dW terms cancel (the local diffusion risk is hedged), and the portfolio becomes locally risk-free:

$$d\Pi = - \left(V_t + \frac{1}{2} \sigma^2(t)S^{2\gamma} V_{SS} \right) dt.$$

Under no-arbitrage in the risk-neutral world, a risk-free portfolio must earn the risk-free rate:

$$d\Pi = r\Pi dt = r(-V + SV_S) dt.$$

Equating these and rearranging yields the pricing PDE:

$$V_t + rSV_S + \frac{1}{2} \sigma^2(t)S^{2\gamma} V_{SS} - rV = 0,$$

with terminal condition for a European put:

$$V(S, T) = (K - S)^+.$$

This PDE is what we discretise in the Crank–Nicolson method, and it also links to Monte Carlo via Feynman–Kac.

For the market-data application (Part B), the same hedging logic produces the standard Black–Scholes PDE under GBM, with the minor modification that a continuous dividend yield q enters the drift and boundary behaviour. In that setting we use Black–Scholes as a baseline model, Monte Carlo under GBM as a probabilistic cross-check, and Crank–Nicolson as the PDE-based solver.

3.2 Monte Carlo simulation

By Feynman–Kac, the put price can be written as a discounted expectation:

$$V(S, t) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} (K - S_T)^+ \mid S_t = S \right].$$

Monte Carlo approximates this expectation by simulating many paths of S_t and averaging the discounted payoff.

The law of large numbers justifies averaging independent discounted payoffs. With finite payoff variance, the central limit theorem gives sampling error of order $\mathcal{O}(N^{-1/2})$. At a fixed Euler time step, this convergence is to the discretised simulation expectation; increasing the path count alone does not remove time-discretisation bias.

To simulate paths under the risk-neutral SDE, we discretise using Euler–Maruyama with $\Delta t = T/n$:

$$S_{i+1} = S_i + r S_i \Delta t + \sigma(t_i) S_i^\gamma \sqrt{\Delta t} \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 1).$$

In the computations we used $n = 100$ time steps and $N = 10,000$ simulated paths.

The supplied Euler implementation uses $\max(S, 0)$ inside the fractional diffusion power, while the price update itself is not constrained to be non-negative. This boundary treatment is part of the numerical approximation and should be examined in refinement checks.

3.2.1 Price and sampling interval

The Monte Carlo estimator is:

$$\hat{V} = e^{-rT} \frac{1}{N} \sum_{j=1}^N (K - S_T^{(j)})^+.$$

A 95% confidence interval is:

$$\hat{V} \pm z_{0.975} \frac{\hat{s}}{\sqrt{N}},$$

where $\hat{s}$ is the sample standard deviation of discounted payoffs.

This interval describes sampling uncertainty under the simulated model. It excludes discretisation bias, parameter uncertainty and model misspecification.

3.2.2 Delta estimation

Delta is defined as $\Delta = \frac{\partial V}{\partial S}$. We approximate it using a central difference:

$$\Delta \approx \frac{V(S_0 + \epsilon) - V(S_0 - \epsilon)}{2\epsilon},$$

with $\epsilon = 0.01 S_0$. To reduce variance, we used common random numbers when estimating $V(S_0 + \epsilon)$ and $V(S_0 - \epsilon)$.

3.3 Crank–Nicolson finite differences

The Crank–Nicolson method blends explicit and implicit schemes and takes their average. In practice, this gives a second-order accurate time discretisation (in the smooth regime) and is typically much more stable than a purely explicit method.

Like the Monte Carlo method, discretisation is required. However, here we discretise the pricing PDE directly on a grid. We choose a spatial grid

$$S_i = i \delta S, \quad i = 0, 1, \dots, M,$$

and a time grid

$$t_j = j \delta t, \quad j = 0, 1, \dots, P, \quad \delta t = \frac{T}{P}.$$

We write $V_i(t) \approx V(S_i, t)$, and at each fixed t we apply central differences in S :

$$V_S(S_i, t) \approx \frac{V_{i+1}(t) - V_{i-1}(t)}{2 \delta S}, \quad V_{SS}(S_i, t) \approx \frac{V_{i+1}(t) - 2V_i(t) + V_{i-1}(t)}{(\delta S)^2}.$$

Substituting these into the PDE and rearranging to isolate the time derivative gives a semi-discrete ODE system for each interior node $i = 1, \dots, M - 1$:

$$\frac{dV_i}{dt} + \frac{1}{2} \sigma^2(t) (S_i)^{2\gamma} \frac{V_{i+1} - 2V_i + V_{i-1}}{(\delta S)^2} + r S_i \frac{V_{i+1} - V_{i-1}}{2 \delta S} - r V_i = 0.$$

Equivalently,

$$\frac{dV_i}{dt} = a_i(t) V_{i-1} + b_i(t) V_i + c_i(t) V_{i+1},$$

where the coefficients are:

$$\begin{aligned} a_i(t) &= -\frac{1}{2} \frac{\sigma^2(t) (S_i)^{2\gamma}}{(\delta S)^2} + \frac{r S_i}{2 \delta S} \\ b_i(t) &= r + \frac{\sigma^2(t) (S_i)^{2\gamma}}{(\delta S)^2} \\ c_i(t) &= -\frac{1}{2} \frac{\sigma^2(t) (S_i)^{2\gamma}}{(\delta S)^2} - \frac{r S_i}{2 \delta S} \end{aligned}$$

Let $\mathbf{V}(t)$ denote the vector of interior option values

$$\mathbf{V}(t) = (V_1(t), V_2(t), \dots, V_{M-1}(t))^{\top}.$$

Stacking the semi-discrete equations across all interior nodes yields a linear ODE system of the form:

$$\frac{d\mathbf{V}}{dt} = \Lambda(t) \mathbf{V} + \mathbf{w}(t),$$

where:

$\mathbf{V}(t)$ is the vector of option values at the interior grid asset prices.

$\Lambda(t)$ is the tridiagonal matrix containing the drift and diffusion coefficients $a_i(t), b_i(t), c_i(t)$.

$\mathbf{w}(t)$ is the vector of boundary contributions (only the first and last interior equations depend on the boundary values).

Grid: $(P + 1) \times (M + 1)$ for the full grid, with $(M - 1)$ interior spatial nodes and $(P + 1)$ time levels; $\delta t = T/P$ and $\delta S = S_{\max}/M$.

$$\Lambda(t) = \begin{pmatrix} b_1(t) & c_1(t) & 0 & 0 & \cdots & 0 \\ a_2(t) & b_2(t) & c_2(t) & 0 & \cdots & 0 \\ 0 & a_3(t) & b_3(t) & c_3(t) & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & b_{M-2}(t) & c_{M-2}(t) \\ 0 & 0 & \cdots & 0 & a_{M-1}(t) & b_{M-1}(t) \end{pmatrix}.$$

$$\mathbf{w}(t) = \begin{pmatrix} a_1(t) V_0(t) \\ 0 \\ \vdots \\ 0 \\ c_{M-1}(t) V_M(t) \end{pmatrix},$$

where $V_0(t) = V(0, t)$ and $V_M(t) = V(S_M, t)$ are the boundary values at $S = 0$ and $S = S_{\max}$, respectively. This form makes it clear that the spatial discretisation leads to a tridiagonal linear operator, which is why each Crank–Nicolson time step can be computed efficiently by solving a tridiagonal linear system.

3.3.1 Boundary and terminal conditions

To make the PDE numerically solvable we truncate the spatial domain to $S \in [0, S_{\max}]$, with $S_{\max}$ chosen sufficiently large so that the put is essentially worthless at the top boundary. In the implementation used here we set $S_{\max} = 250$, which is well above $K = 140$ in the reported run.

The boundary and terminal conditions used were:

Top boundary: $V(S_{\max}, t) = 0$.

Bottom boundary: $V(0, t) = Ke^{-r(T-t)}$.

Terminal condition: $V(S, T) = (K - S)^+$.

3.3.2 Time stepping

Crank–Nicolson applies the trapezium rule in time. Writing $\mathbf{V}^j \approx \mathbf{V}(t_j)$, a standard CN discretisation is:

$$\frac{\mathbf{V}^{j+1} - \mathbf{V}^j}{\delta t} = \frac{1}{2} \left(\Lambda^{j+1} \mathbf{V}^{j+1} + \Lambda^j \mathbf{V}^j \right) + \frac{1}{2} (\mathbf{w}^{j+1} + \mathbf{w}^j).$$

Since we solve backwards from the terminal condition at $t = T$, at each step we know $\mathbf{V}^{j+1}$ and solve a tridiagonal linear system to obtain $\mathbf{V}^j$.

The formula above is the time-centred scheme. In the supplied CEV implementation, the time-dependent coefficients are held at the earlier grid time within each step. This implementation detail matters when interpreting the observed temporal convergence; the numerical experiment should not be described as a demonstrated second-order time result.

3.3.3 Rannacher smoothing

The payoff $(K - S)^+$ has a kink at the strike. Rannacher smoothing replaces each of the first two time intervals nearest maturity with two backward-Euler half-steps, giving four half-steps before the Crank–Nicolson updates. This damps initial oscillations from the non-smooth payoff.

3.3.4 Price and delta extraction

For the reported Part A solution, the grid uses $P = 100$ time steps and $M = 300$ spatial steps on $S \in [0, 250]$. The price is interpolated to $S_0 = 140$ from the $t = 0$ grid. Delta is calculated with the central difference

$$\Delta(S_i, 0) \approx \frac{V_{i+1}^0 - V_{i-1}^0}{2\delta S},$$

and interpolated to the same spot value.

4 Results and discussion

4.1 Part A: price and delta comparison

Since the model is not Black–Scholes, the closed-form Black–Scholes price is not an exact ground truth. However, we can construct an estimated benchmark using an effective volatility:

$$\sigma_{\text{eff}} \approx \left(S_0^{2(\gamma-1)} \frac{1}{T} \int_0^T \sigma(t)^2 dt \right)^{1/2}.$$

This gave $\sigma_{\text{eff}} = 0.1687$ and the estimated Black–Scholes put price 9.1492. Both numerical prices are close to this baseline, and the Black–Scholes value lies inside the Monte Carlo confidence interval.

Table 1. Initial price and delta comparison (4 d.p.).

Quantity	Value
Black–Scholes (estimated) put price	9.1492
Effective volatility σ_{eff}	0.1687
Monte Carlo put price	9.0577
95% confidence interval (price)	[8.7989, 9.3165]
Monte Carlo delta	-0.4009
95% confidence interval (delta)	[-0.4091, -0.3927]
Crank–Nicolson put price	9.1295
Crank–Nicolson delta	-0.4071

Table 1 provides an initial consistency check. Both the Crank–Nicolson price and the approximate Black–Scholes reference lie within the Monte Carlo price interval. Agreement at this precision supports consistency between the calculations, while leaving discretisation and approximation errors to be assessed separately.

The Crank–Nicolson delta also lies inside the Monte Carlo 95% interval. This supports consistency of the hedge-ratio estimates at S_0 , without establishing accuracy across all grid points or parameter settings.

4.2 Additional sensitivities

Using finite-difference bumps (as implemented in the code), we obtained:

$$\Gamma \approx 0.0141, \quad \text{Vega} \approx 49.2514, \quad \Theta \approx -3.8189, \quad \rho \approx -91.0930.$$

Vega and rho are reported per unit change in decimal volatility and interest rate. For example, $\rho \approx -91.09$ implies a first-order price change of approximately -0.91 for a one-percentage-point increase in r . These are local finite-difference estimates, so bump size and simulation variance affect their precision.

4.3 Solution and hedge-profile diagnostics

The path band illustrates dispersion in simulated underlying prices. It is distinct from a confidence interval for the estimated option price and does not validate pricing accuracy.

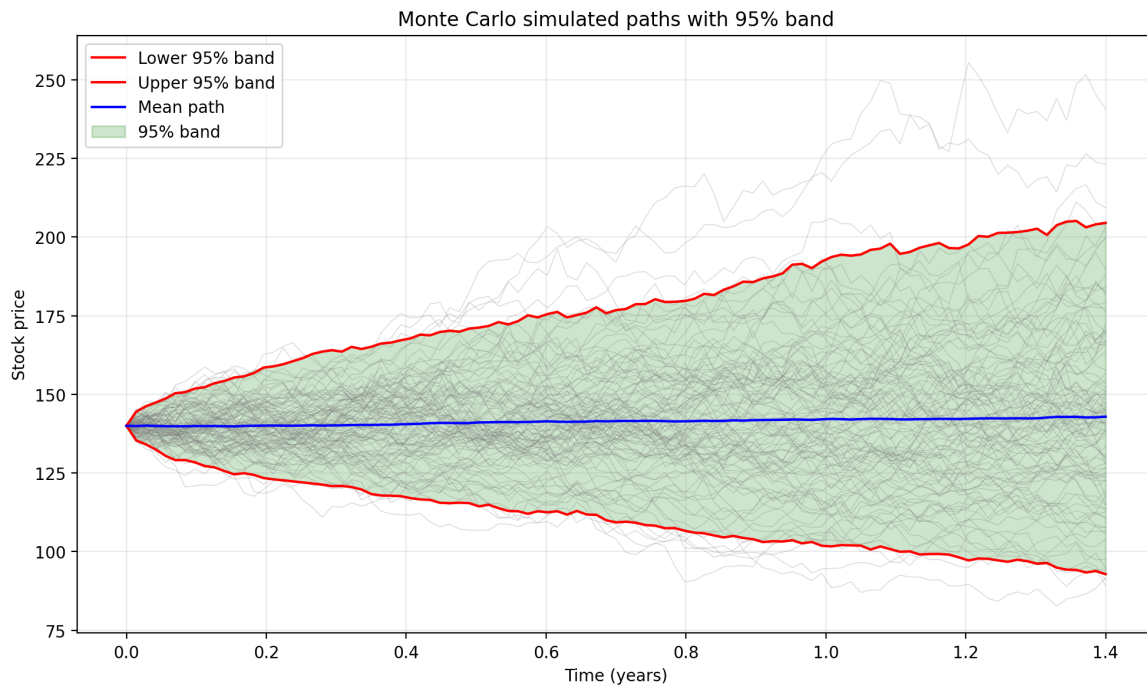


Figure 1. Monte Carlo simulated stock paths with 95% band and mean path.

The value surface shows the numerical solution approaching the terminal payoff. The delta profile moves from approximately -1 at low spot values towards zero at high spot values, consistent with the qualitative shape of a put. These are useful shape checks, not independent proofs of numerical accuracy.

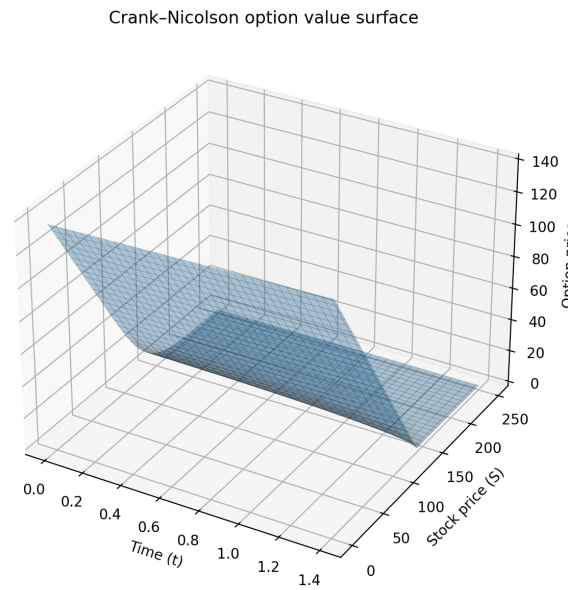


Figure 2. Crank–Nicolson value surface $V(S, t)$ over $S \in [0, 250]$ and $t \in [0, T]$.

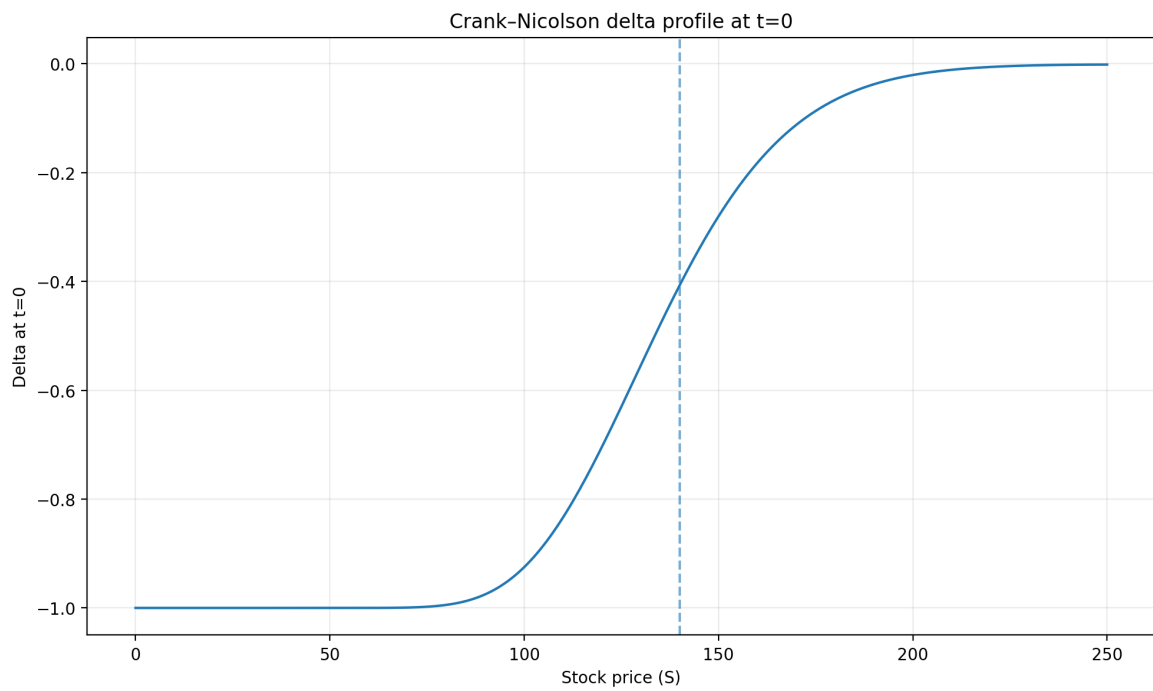


Figure 3. Crank–Nicolson delta profile $\Delta(S, 0)$ across the spatial grid (dashed line at $S = K$).

4.4 Convergence and sampling error

To compare numerical behaviour more directly, we plotted relative errors against high-resolution reference values:

- Monte Carlo errors are computed against a higher-order Monte Carlo estimate using $N_{\text{ref}} = 1,000,000$ paths (generated independently).
- Crank–Nicolson spatial errors fix $P = 500$ and compare against a high-resolution spatial reference with $M_{\text{ref}} = 100,000$.
- Crank–Nicolson temporal errors fix $M = 500$ and compare against a high-resolution temporal reference with $P_{\text{ref}} = 100,000$.

The horizontal axis uses N for Monte Carlo and $M \cdot P$ for Crank–Nicolson. These are refinement proxies, not matched runtimes or operation counts: Monte Carlo path work also depends on the number of time steps.

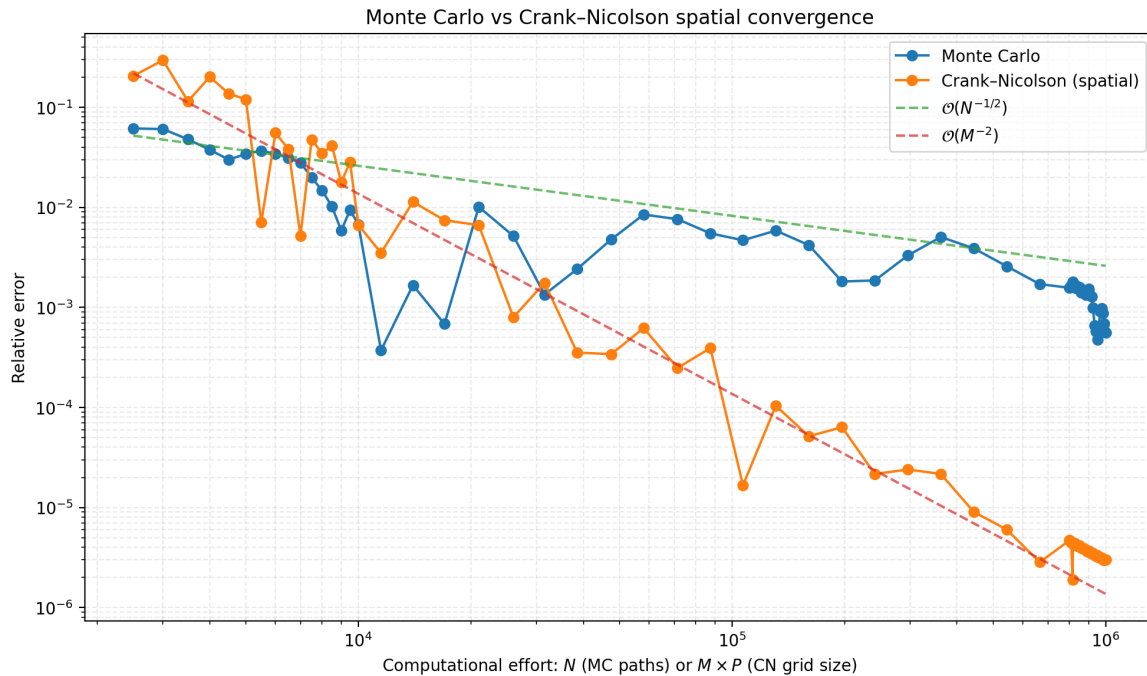


Figure 4. Monte Carlo relative error vs spatial Crank–Nicolson relative error (both measured against high-resolution references).

Figures 4 and 5 are the *primary* numerical comparison in Part A. They show a clear separation between statistical error (Monte Carlo) and discretisation error (Crank–Nicolson). On the right of both plots, at effort around 10^6 , the Monte Carlo relative error sits around the 10^{-3} scale (a few $\times 10^{-4}$ to 10^{-3}), consistent with the slow $\mathcal{O}(N^{-1/2})$ rate.

In contrast, the Crank–Nicolson curves fall much faster. In the spatial refinement plot (Figure 4), Crank–Nicolson reaches around 10^{-5} error already by effort on the order of a few $\times 10^5$, and pushes into the few $\times 10^{-6}$ range by 10^6 , aligning with the $\mathcal{O}(M^{-2})$ guide once the asymptotic regime is reached. The early “spikiness” is expected before the clean second-order regime dominates (and is also influenced by reference-value accuracy and the payoff kink).

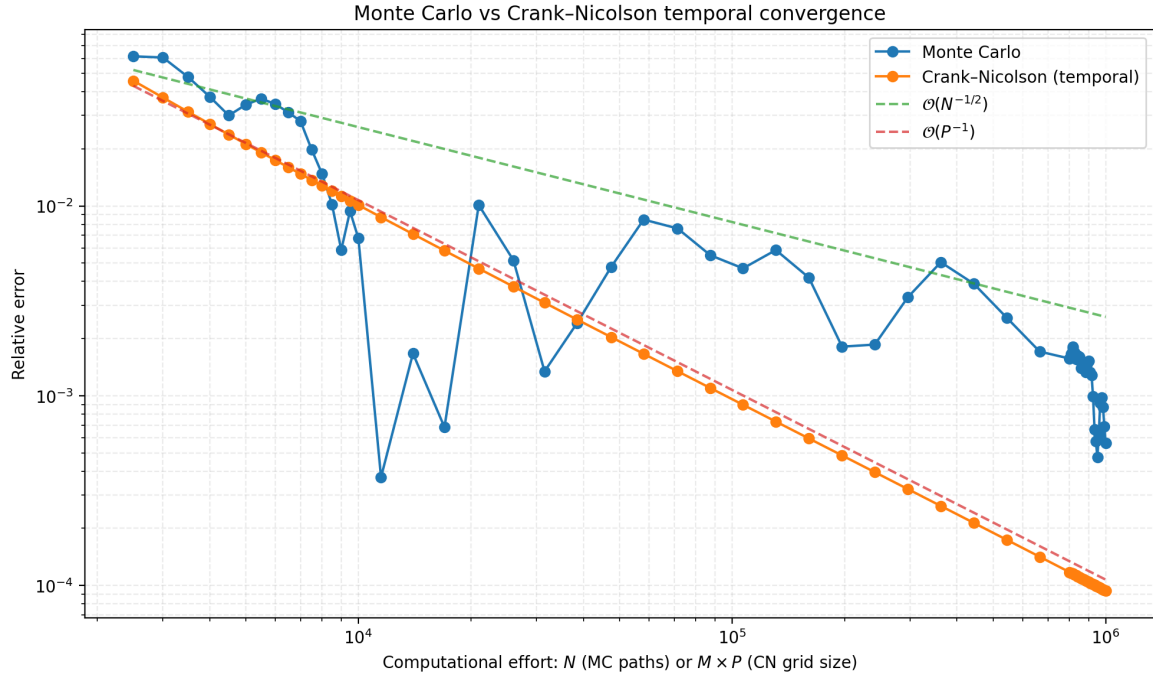


Figure 5. Monte Carlo relative error vs temporal Crank–Nicolson relative error (both measured against high-resolution references).

For temporal refinement (Figure 5), the observed error follows approximately the $\mathcal{O}(P^{-1})$ guide and reaches around 10^{-4} at the plotted effort of 10^6 . The experiment therefore shows slower temporal than spatial refinement. The time treatment of the coefficients, non-smooth payoff and reference error all warrant examination before attributing this behaviour to a single cause. These plots compare convergence patterns; they do not establish a runtime-normalised efficiency ranking.

Figure 6 isolates the sampling-error scale, $\hat{\sigma}/\sqrt{N}$. Its $N^{-1/2}$ slope follows directly from this formula and is not an independent empirical validation of the pricing model.

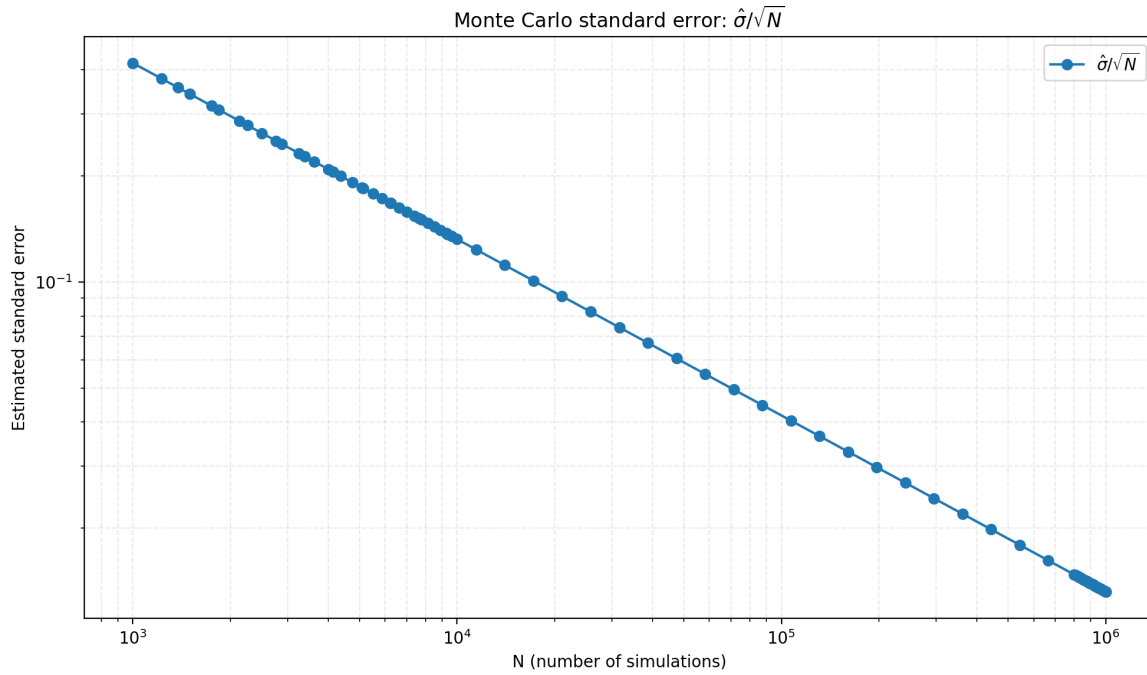


Figure 6. Monte Carlo estimated standard error line $\hat{\sigma}/\sqrt{N}$ (log–log), which follows an $N^{-1/2}$ slope by construction.

4.5 Part B: option-chain comparison

Part B applies European pricing calculations to the supplied SPY chain under GBM with continuous dividend yield q . For each contract it reports Black–Scholes prices using the supplied implied volatility, MC prices and sampling intervals, CN prices, Black–Scholes Greeks, and model–mid differences.

The run used an SPY spot level of 682.85 and a dividend yield $q = 1.050\%$. The cleaned chain contained 947 contracts across 10 expiries and strikes from 510 to 750.

Table 2. SPY chain summary (Part B).

Measure	Value
Options priced	947
Calls / Puts	280 / 667
Strike range	510 – 750
Implied vol range (mean)	10.7% – 53.5% (22.48%)
Dividend yield q	1.050%
Market mid mean (std)	3.2851 (3.1712)
BS mean (std)	3.0320 (3.0822)
MC mean (std)	2.9835 (3.0542)
CN mean (std)	3.0282 (3.0689)

The BS and CN aggregate prices in Table 2 are close. This is a useful internal consistency check because both use the same model assumptions. Aggregate agreement can hide contract-level differences, so it is not sufficient validation by itself. The lower MC mean should be investigated

against contract-level sampling errors and common inputs rather than attributed to finite sample size without a test.

Table 3. Model–mid difference metrics vs market mid (Part B).

Model	Mean (model-mid)	MAE	MaxAbs
BS	-0.2531	0.3065	1.0717
MC	-0.3016	0.3378	1.1653
CN	-0.2569	0.3102	1.1130

Table 3 reports model prices minus quoted mids. These differences are diagnostics, not evidence of executable arbitrage. Possible contributors include bid–ask effects, input or timestamp mismatches, volatility-field conventions and the European pricing assumptions. Similar BS and CN averages support numerical consistency, but cannot exclude shared input or model errors.

4.6 Implied volatility and risk structure

For the SPY chain, the implied volatility skew and term structure provide immediate context for risk. Figure 7 shows the skew across expiries, while Figure 8 summarises how Delta/Gamma/Vega exposures vary by strike and maturity.

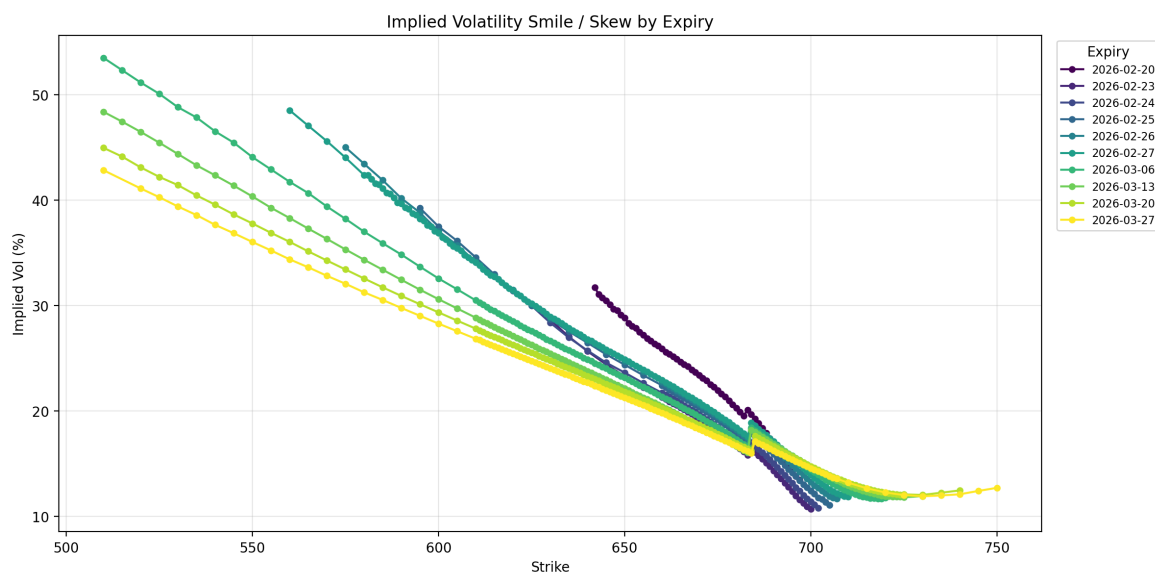


Figure 7. SPY implied volatility smile/skew by expiry (Part B).

Figure 7 shows a pronounced *downside skew*: implied volatilities are substantially higher at lower strikes (deep OTM puts) and decline toward the ATM region. Across the higher-strike region the curves flatten and even show a slight right-tail uptick for some maturities, consistent with a skew/smile shape rather than a perfectly flat volatility surface. The separation between expiry curves reflects term structure: overall level and slope vary by maturity, which is precisely why a chain-level engine needs to handle expiry-by-expiry inputs rather than a single volatility number.

Figure 8 matches standard option-risk intuition and also acts as a sign/convention check on the engine outputs. Gamma peaks around the ATM region (near spot), with the highest peaks occurring in the shorter-dated expiries, while longer-dated expiries exhibit lower gamma but higher vega. The

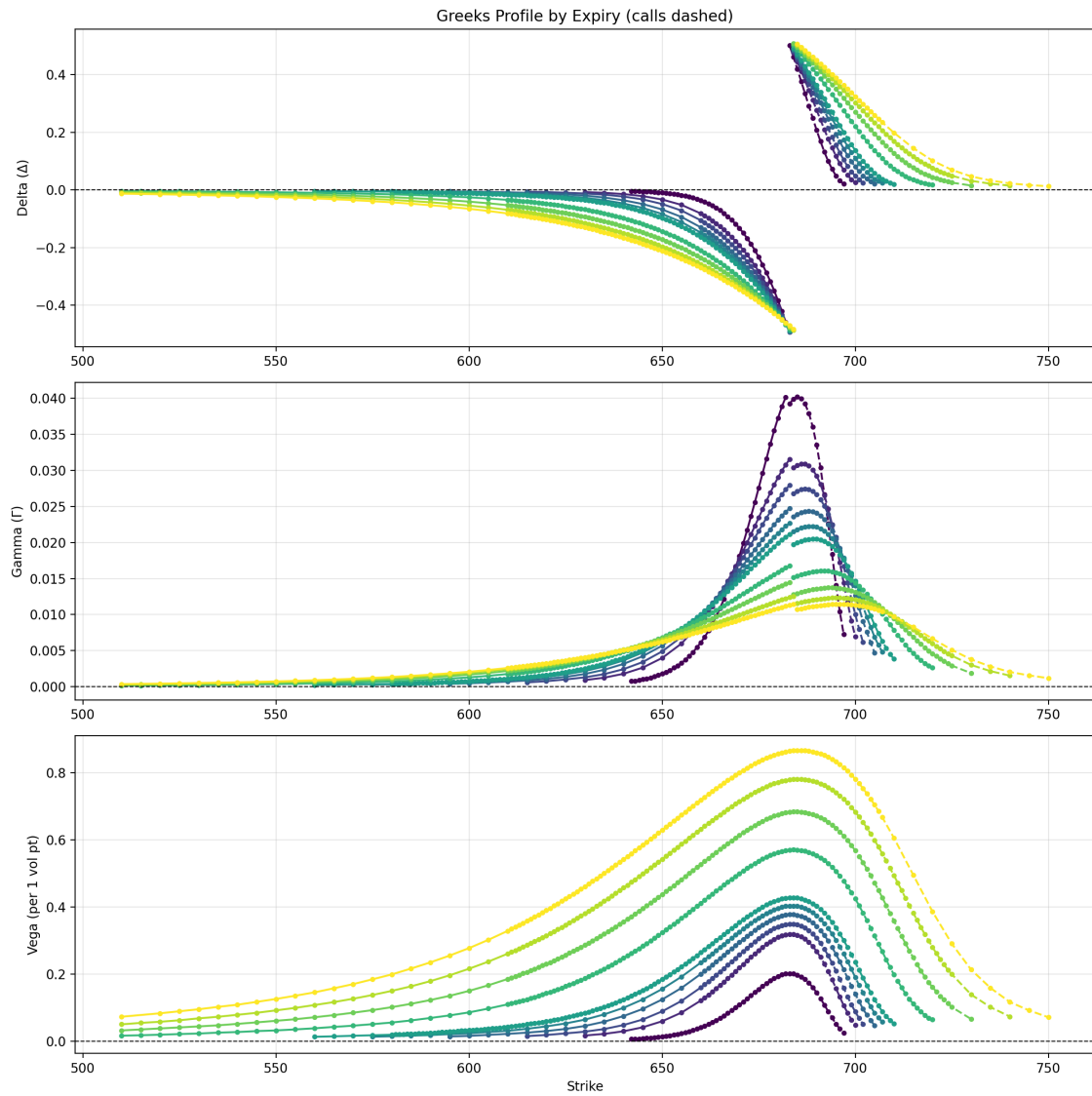


Figure 8. SPY Greeks profiles by expiry (Black–Scholes Greeks; calls dashed) (Part B).

vega curves rise with maturity and peak around ATM, reflecting greater sensitivity of longer-dated options to volatility changes.

4.7 Model–market diagnostics

Figure 9 shows negative model–mid differences for all three methods and similar BS and CN distributions. The retained plot annotates the MC mean at approximately -0.27 , whereas Table 3 reports -0.3016 . That difference between the supplied figure and table must be reconciled by regenerating both from the same recorded run before using them as a single quantitative validation result.

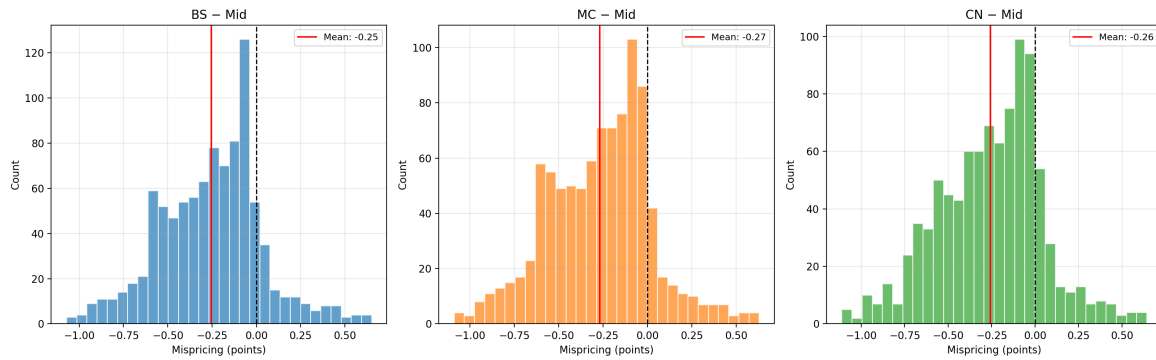


Figure 9. Mispricing distributions (model minus market mid) for BS, MC and CN (Part B).

For the near-the-money subset in Figure 10, several market mids lie above the MC sampling intervals. This suggests that sampling uncertainty alone is insufficient to explain the differences for those contracts. Input alignment and contract conventions should be checked before assigning the discrepancy to any particular cause.

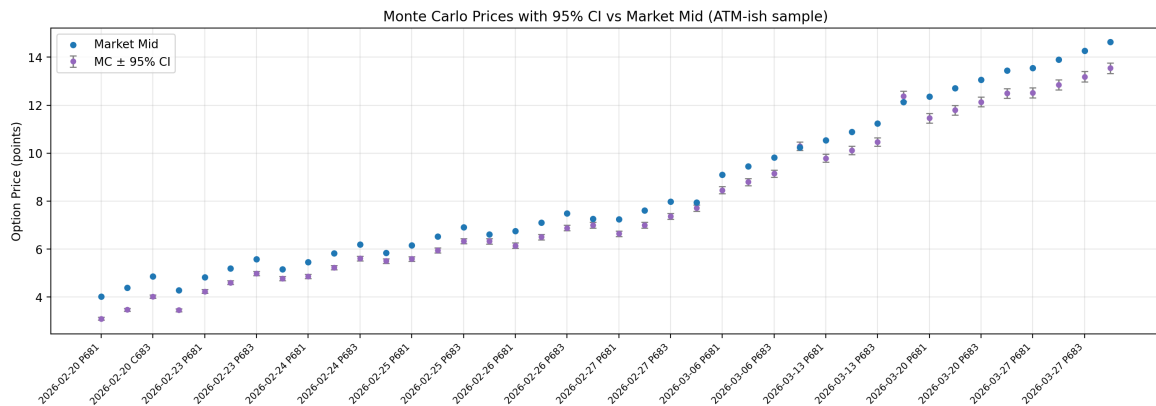


Figure 10. Monte Carlo price estimates with 95% CI vs market mid for an near-the-money subset (Part B).

4.8 Scenario price changes

The scenario analysis selects six contracts nearest to the money and reprices them under spot shocks of $\{-10\%, -5\%, 0\%, +5\%, +10\%\}$ and volatility shocks of $\{-5\text{pp}, -2\text{pp}, 0, +2\text{pp}, +5\text{pp}\}$. Here pp denotes absolute percentage points of implied volatility. The plotted value is the sum of price changes across the selected contracts, with one unit of each option and no contract multiplier.

Figure 11 illustrates the joint spot and volatility shocks, but its zero-spot/zero-volatility cell is nonzero. Since the chart is intended to show changes from the base value, that residual indicates a

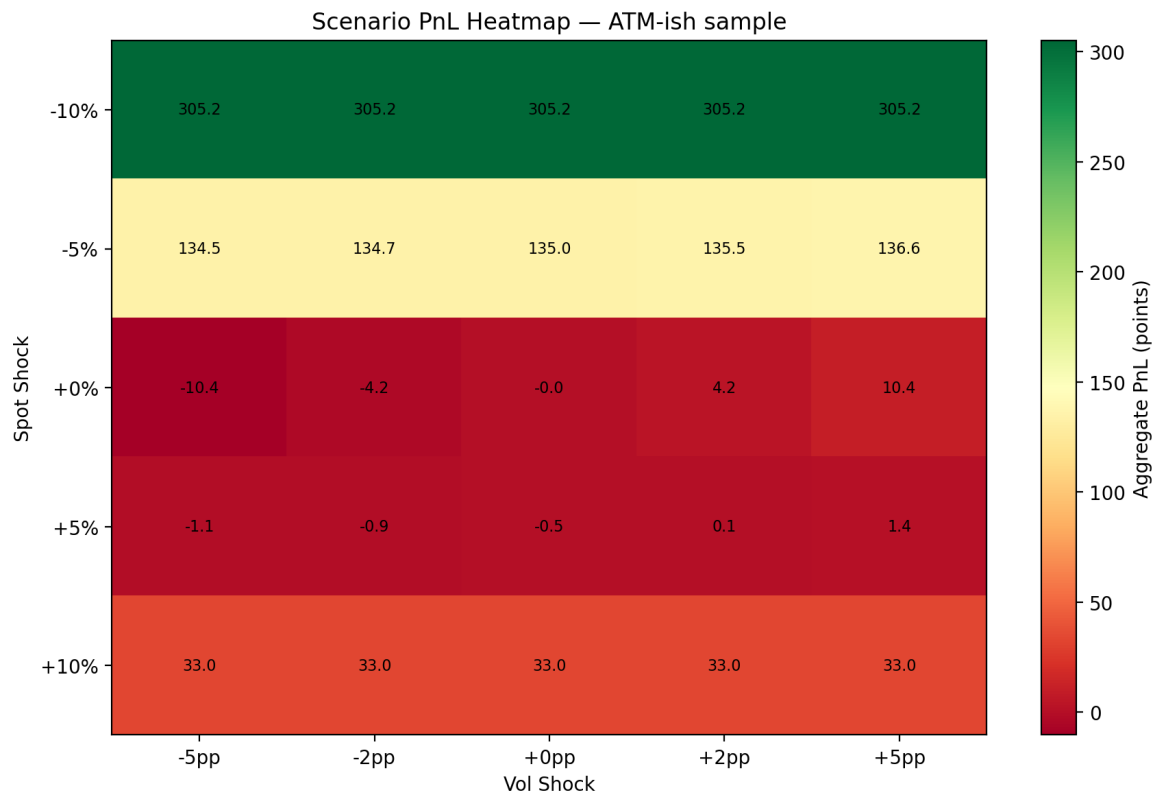


Figure 11. Scenario PnL heatmap for an near-the-money subset under joint spot and volatility shocks (sum of model price changes in quote units).

mismatch between the repricing calculation and its reference prices. The figure is retained as an output requiring reconciliation, not as a validated PnL result.

The next check is to align option-type labels, input conventions and base prices between the pricing engine and scenario routine, then regenerate the heatmap. Once reconciled, finite shocks would complement the local Greek sensitivities.

5 Limitations and further work

- **Approximate reference in Part A.** The effective-volatility Black–Scholes price is not an exact CEV solution. Numerical agreement with it does not establish convergence to the true diffusion price.
- **Simulation and grid error.** MC intervals cover sampling error only. Euler boundary treatment, time steps, finite-difference domain truncation and Greek bump sizes require separate refinement checks.
- **Temporal discretisation.** The supplied CEV code holds time-dependent coefficients fixed within a step. The reported temporal experiment is approximately first order; it should not be presented as a verified second-order result.
- **Market assumptions.** The Part B calculations use European GBM pricing with continuous dividends. Exercise rights, discrete dividends, settlement and quote timing must be reconciled with the supplied SPY contracts before interpreting model–mid gaps as fair-value errors.

- **Run consistency.** The MC mean shown in the supplied distribution plot differs from the summary table. The values and figure are retained, with the discrepancy identified explicitly; a reproducible rerun should regenerate all outputs together.
- **Scenario scope and consistency.** The heatmap sums changes in model prices for a small subset. Its zero-shock cell is nonzero and needs reconciliation with the base prices. It also excludes actual position sizes, contract multipliers, transaction costs and dynamic hedging, so it is not a realised trading PnL.

Further work should record a common input snapshot and random seed, compare contract-level numerical errors, refine time and space grids independently, and benchmark runtime directly. Market comparisons should first align the contract and data conventions with the chosen pricing model.

6 Conclusion

Monte Carlo and Crank–Nicolson produce compatible Part A prices and deltas at the reported precision. The comparison illustrates the distinction between sampling error and discretisation error, while the sensitivity and convergence checks identify where further numerical validation is needed.

The option-chain application extends the methods into a repeatable pricing and risk-reporting workflow. Similar BS and CN prices provide an internal consistency check; market-mid differences highlight the importance of input and contract conventions. Together, the results demonstrate numerical implementation and diagnostic reasoning without claiming independent market calibration or trading profitability.

References

- [1] John Armstrong. Lecture notes for 7CCMFM06, Computational and Numerical Methods in Mathematical Finance. <https://keats.kcl.ac.uk/course/view.php?id=119832>, 2025.